Framework Presentation and Lab Setup Proposal

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INESC-TEC B2.1 Room

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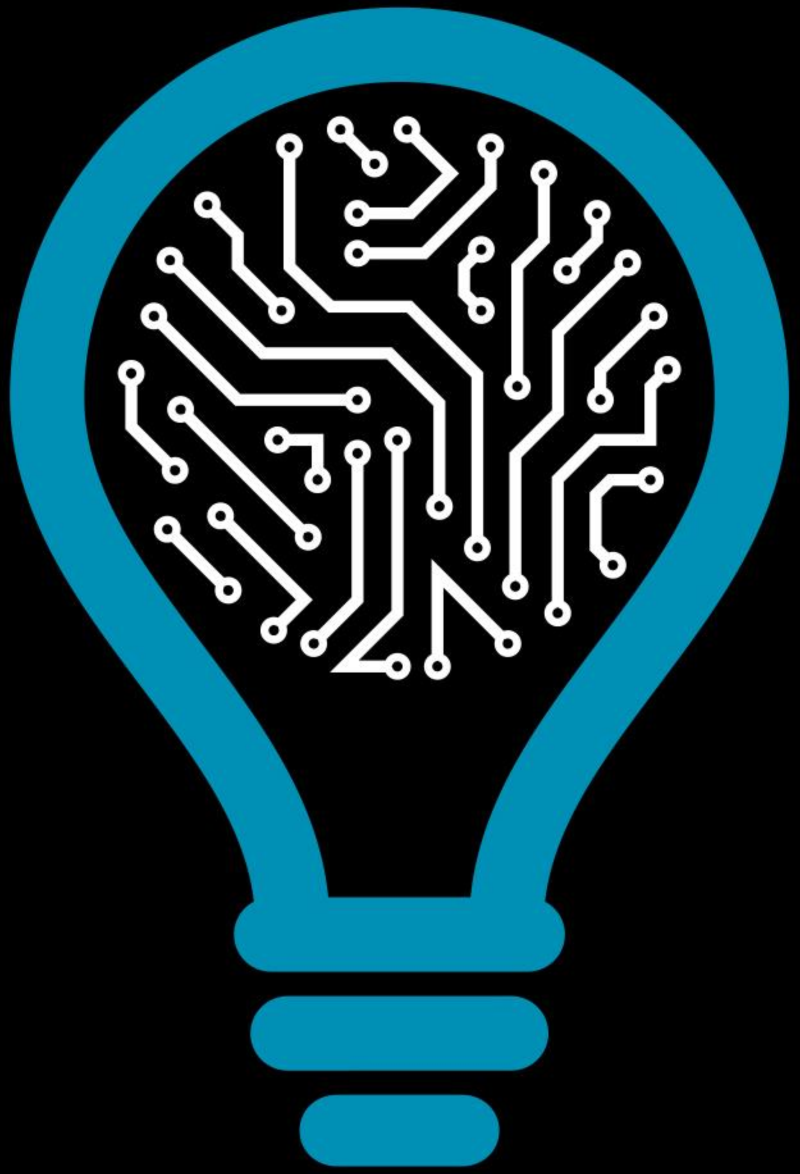

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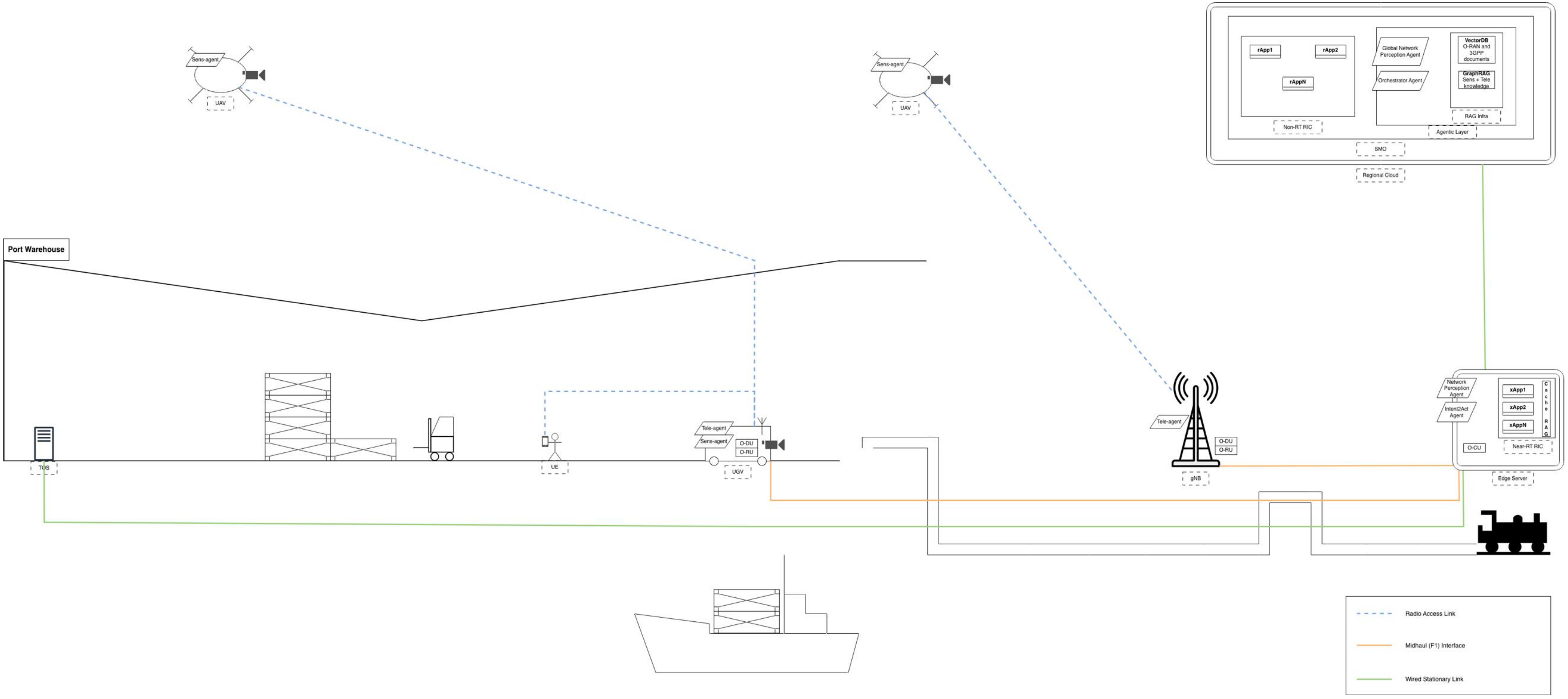
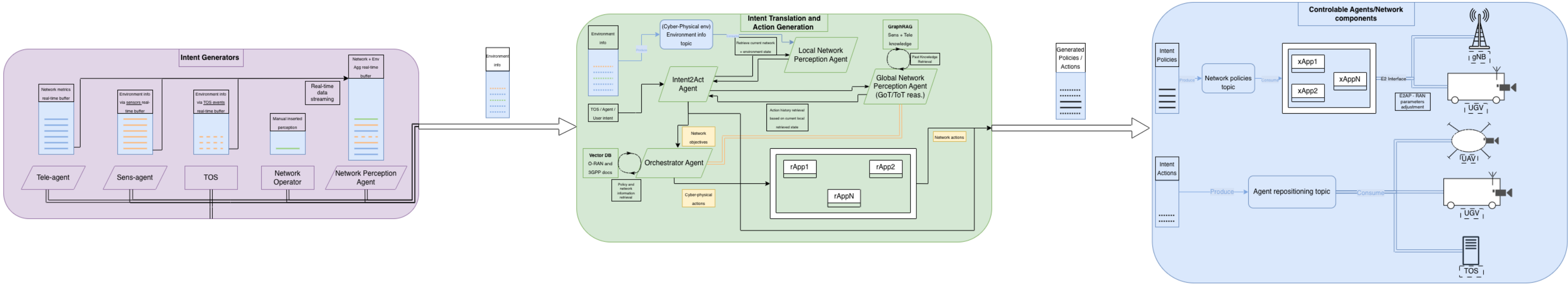
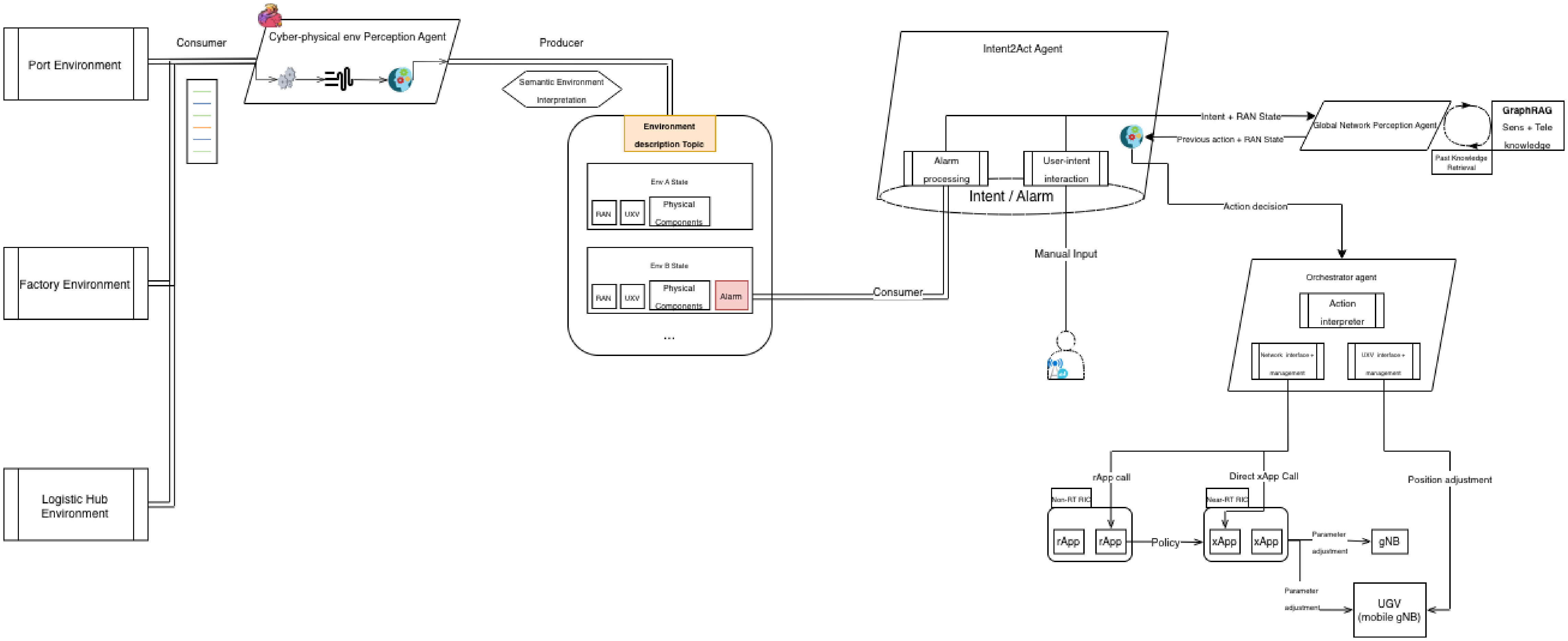
Physical Scenario Presentation – Visual Description



Agents and Network Elements Interaction – A first proposal and key components



Agents Interaction and Orchestration – Pipeline of coordinated intent-retrieval-reasoning-orchestration flow



Future iteractions

- **Lab Setup Architecture CI -** While performing and building the lab setup it is important to improve the architecture "First Lab Setup" and include all the used resources (hardware), all the custom and open-source interfaces, data flow and input/output of the interactions.

- **Expression Mechanism Description** – Generate equations and mathematical expressions to explain and describe the interactions and the improvements of this approach. Describe better the offline and online inference processing.

- **Definition of intents** - Consider to limit agents (via API) and users (via UI) intent to relief the paper of intent translation/ decoding.

- **Multi-agent debate and personas** – Create streaming based environment to create debate between the agents (if needed) selecting and testing multiple personas.

- **Clustering of Solutions** – Ground the usage of big data applications and sustain its relevance to O-RAN architetures and B5G deployments.
    - <u>Interoperability and multi-vendor information exchange</u> – Describe, based on the proposed architecture, how the vendors could contribute with their knowledge to a "global" inference mechanism using RAG/GraphRAG without exposing network information, metrics, and contributing to a global network of knowledge.
    - <u>Solutions deployment</u> – The agents deployment should be explored. Our solution has some potential to employ edge LLMs that will allow, for exemple, the UEs (or even the UXVs), to express some intent or critical noticed modification o n the environment or network.

- **Knowledge Acquirement and Reasoning Experiments** – Based on the literature there are some promising mechanisms of reasoning, data retrieval, and local-KB + cache-KB that could be adapted to our scenario. Some experiments during the process of prompting, reasoning and knowledge database build and retrieval that could be compared.

Recommended Literature (email attached)

- **Zhu2026** – "Wireless large AI model: shaping the AI-native future of 6G and beyond" (Survey)
- **Jiang2026** – "From Large AI Models to Agentic AI: A Tutorial on Future Intelligent Communications"
- **Barros2025** – "RAN Cortex: Memory-Augmented Intelligence for Context-Aware Decision-Making in AI-Native Networks"

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